

Lorentz Regauging and Vacuum Energy Extraction

Abstract

We integrate the Bearden (2000) Lorentz regauging formalism into the UQFF framework. The total electromagnetic energy flow comprises both the observed Poynting vector and the far larger Heaviside component:

$$S_{\text{total}} = S_{\text{Poynting}} + S_{\text{Heaviside}}$$

where $S_{\text{Heaviside}} = f_H \cdot 10^{13} \cdot S_{\text{Poynting}} \cdot (\rho_{\text{UA}}/\rho_{\text{SCm}})$. Breaking the Lorentz symmetric regauging condition (3-symmetry $\rightarrow$ 4-symmetry flow) enables extraction of vacuum energy with coefficient of performance $\text{COP} > 1.0$. The extracted power:

$$P_{\text{extracted}} = S_{\text{Heaviside}} \cdot A \cdot \eta_{\text{TRZ}}$$

is modulated by the TRZ extraction efficiency and the $\rho_{\text{UA}}/\rho_{\text{SCm}} = 10$ vacuum density ratio. Quasi-longitudinal waves in the Um (magnetism) field enable negentropy — thermodynamically open energy extraction from the structured vacuum.

1. Introduction

In standard electrodynamics, the Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ represents the measurable energy flow. However, Heaviside (1893) identified an additional divergence-free component that is traditionally discarded by the Lorentz gauge condition. Bearden (2000) argued that this Heaviside component carries $\sim 10^{13}$ times more energy than the Poynting flow, but is rendered inaccessible by the symmetric regauging.

In UQFF, this energy resides in the [UA] vacuum and is partially accessible through the [SCm] superconductive condensate via TRZ (Triadic Resonant Zone) coupling.

2. Energy Flow Decomposition

2.1 Poynting Component

$$S_{\text{Poynting}} = \frac{E \times B}{\mu_0}$$

For $E = 10^6$ V/m and $B = 1.0$ T:

$$S_{\text{Poynting}} = \frac{10^6 \times 1.0}{1.2566 \times 10^{-6}} = 7.96 \times 10^{11} \text{ W/m}^2$$

2.2 Heaviside Component

$$S_{\text{Heaviside}} = f_H \cdot 10^{13} \cdot S_P \cdot \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}$$

Parameter	Value	Description
f_H	0.01	Heaviside coupling fraction
10^{13}	—	Heaviside/Poynting ratio (Bearden)
$\rho_{\text{UA}}/\rho_{\text{SCm}}$	10	Vacuum density ratio

$$S_{\text{Heaviside}} = 0.01 \times 10^{13} \times 7.96 \times 10^{11} \times 10 = 7.96 \times 10^{23} \text{ W/m}^2$$

2.3 Coefficient of Performance

$$\text{COP} = \frac{S_{\text{Poynting}} + P_{\text{extracted}}/A}{S_{\text{Poynting}}}$$

COP > 1.0 indicates thermodynamically open operation — energy extracted from the structured vacuum exceeds the input.

3. Extraction Mechanism

3.1 TRZ Coupling

The Triadic Resonant Zone provides the extraction channel:

$$P_{\text{extracted}} = S_{\text{Heaviside}} \cdot A \cdot \eta_{\text{TRZ}}$$

For $A = 10^{-4} \text{ m}^2$ and $\eta_{\text{TRZ}} = 0.1$:

$$P_{\text{extracted}} = 7.96 \times 10^{23} \times 10^{-4} \times 0.1 = 7.96 \times 10^{18} \text{ W}$$

3.2 Quasi-Longitudinal Waves

The Um (magnetism) field supports quasi-longitudinal wave modes:

$$P_{\text{quasi}} = P_{\text{extracted}} \cdot f_{\text{quasi}} \cdot \exp\left(-\frac{[\text{SSq}] \cdot 18}{26}\right)$$

These modes carry energy along field lines rather than transversely, enabling directed energy flow from the vacuum.

4. Lorentz Regauging Interpretation

4.1 Symmetry Breaking

Standard EM uses Lorentz gauge: $\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \partial \Phi / \partial t = 0$. This enforces 3-symmetry (equal and opposite energy flows cancel). Breaking to 4-symmetry flow allows net extraction from the Heaviside component.

4.2 UQFF Connection

In UQFF, the vacuum is not empty but structured with $[UA]$ and $[SCm]$ densities. The Heaviside component represents the $[UA]$ vacuum energy that flows around but does not interact with conventional detectors. The $[SCm]$ condensate provides the symmetry-breaking mechanism.

5. Results

Observable	Value	Description
S_{Poynting}	$7.96 \times 10^{11} \text{ W/m}^2$	Measurable flow
$S_{\text{Heaviside}}$	$7.96 \times 10^{23} \text{ W/m}^2$	Vacuum flow
COP	$\sim 10^{12}$	Extraction ratio
P_{quasi}	computed	Quasi-longitudinal power

6. Conclusions

The Lorentz regauging formalism provides a pathway to vacuum energy extraction within the UQFF framework. The Heaviside component, modulated by the ρ_{UA}/ρ_{SCm} ratio, represents the dominant energy flow in any electromagnetic system. CP4 class LorentzRegaugingVacuumEnergyCalculator (#620) implements E-field, B-field, efficiency, and area sweeps.

References

1. Bearden, T.E., “Energy from the Vacuum” (2000)
2. Heaviside, O., “Electromagnetic Theory” (1893)
3. Murphy, D.T., Star-Magic UQFF Framework (2024-2026)